EXAM PAPERS PRACTICE

Q1.

Marking Instructions	AO	Marks	Typical Solution
Selects suitable split in choosing u and v using integration by parts	AO3.1a	M1	$I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ $= \int_0^{\frac{\pi}{2}} (\sin^{n-1} x)(\sin x) dx$
Uses integration by parts (allow one sign error)	AO1.1a	M1	$u = \sin^{n-1} x$
Integrates fully correctly (no need to be simplified)	AO1.1b	A1	$\frac{du}{dx} = (n-1)\sin^{n-2} x \cos x$
Uses the identity $\cos^2 x = 1 - \sin^2 x$ in $\int \cos^2 x \sin^{n-2} x \, dx$ (PI)	AO1.1b	B1	$\frac{dv}{dx} = \sin x \qquad v = -\cos x$ $I_n = \left[-\cos x \sin^{n-1} x\right]_0^{\frac{\pi}{2}}$ $+ (n-1)\int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$
Completes rigorous argument to show result with all steps in the argument clearly set out **AG**	AO2.1	R1	$I_n = [0]$ $+ (n-1)\int_0^{\frac{\pi}{2}} \sin^{n-2} x(1 - \sin^2 x) dx$ $= (n-1)\int_0^{\frac{\pi}{2}} \sin^{n-2} x - \sin^n x \, dx$ $= (n-1)(I_{n-2} - I_n)$ $\therefore I_n = (n-1) I_{n-2} - nI_n + I_n$ $\Rightarrow nI_n = (n-1) I_{n-2}$ **AG**

ALT				$I_n = \int_0^{\frac{\pi}{2}} \sin^{n-2} x \sin^2 x \, dx$ $= \int_0^{\frac{\pi}{2}} \sin^{n-2} x(1 - \cos^2 x) \, dx$ $= I_{n-2} - \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$ $= I_{n-2} - \int_0^{\frac{\pi}{2}} (\sin^{n-2} x \cos x)\cos x \, dx$ $u = \cos x \qquad \frac{du}{dx} = -\sin x$

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				$\frac{dv}{dx} = \sin^{n-2} x \cos x$ $v = \frac{1}{(n-1)} \sin^{n-1} x$ $I_n = I_{n-2} - \left[\frac{1}{(n-1)} \sin^{n-1} x \cos x\right]_0^{\frac{\pi}{2}}$ $- \int_0^{\frac{\pi}{2}} \frac{1}{(n-1)} \sin x \sin^{n-2} x \, dx$ $= I_{n-2} - [0] - \frac{1}{(n-1)} \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ $= I_{n-2} - \frac{1}{(n-1)} I_n$ $\therefore (n-1) I_n = (n-1) I_{n-2} - I_n$ $\Rightarrow nI_n = (n-1)I_{n-2}$ **AG**
	Total 5 marks			